FIRST ECOSYSTEM MEETING

RegenSeq Open Source Ecosystem

Repurposed Sequencers

New Experimental Possibilities

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INTRODUCTION

What the ecosystem is

We take decommissioned sequencers and repurpose them for things besides sequencing, by developing four components.

Control software

Runs automated protocols and talks to the instruments.

Flow cells

Slides you section tissue onto and run through the sequencer.

Protocols

Assays to run on the repurposed instruments.

Image processing

A pipeline to correct, align, segment and measure your images.

INTRODUCTION

How it started

2020

A pandemic project

Decommissioned sequencers sat at the Genome Center, and we wanted to do something fun and useful with them.

FIRST HACKS

Talking to the hardware

We wrote small commands for the instruments and bolted the pumps onto a microscope.

BETA

A scripted workflow

Published a beta version of the software took in a script and automated the wet-lab steps.

TODAY

Version 3

Faster asynchronous architecture with aligned high level control across sequencers

WHY REPURPOSE SEQUENCERS

A sequencer is a fluorescence microscope with fluidics built in

Proven hardware

Already robust in production, with excellent engineering from Illumina.

No custom build

Skip pairing a pricey microscope with custom fluidics, and writing and maintaining the software yourself.

Minimal modification

Only necessary modification is to remove 3 screws from stage.

WHY REPURPOSE SEQUENCERS

The cost

~100×

cheaper per run than commercial systems (our estimate)

Negligible upfront capital cost

You are reusing a sequencer that would otherwise be thrown away. We will help you find one, and a few at the Genome Center are available.

Request a sequencer

regenseq.github.io/community/find-a-sequencer

COMPONENTS · SOFTWARE

Control software

CODE ON GITHUB

github.com/ReGenSeq

Common automation engine

High-level control software that stays compatible across sequencer models.

LOW-LEVEL IMPLEMENTATIONS FOR EACH MODEL

HiSeq 2500

PySeq2500 available

HiSeq 2000

PySeq2501 available from Goff Lab (JHU)

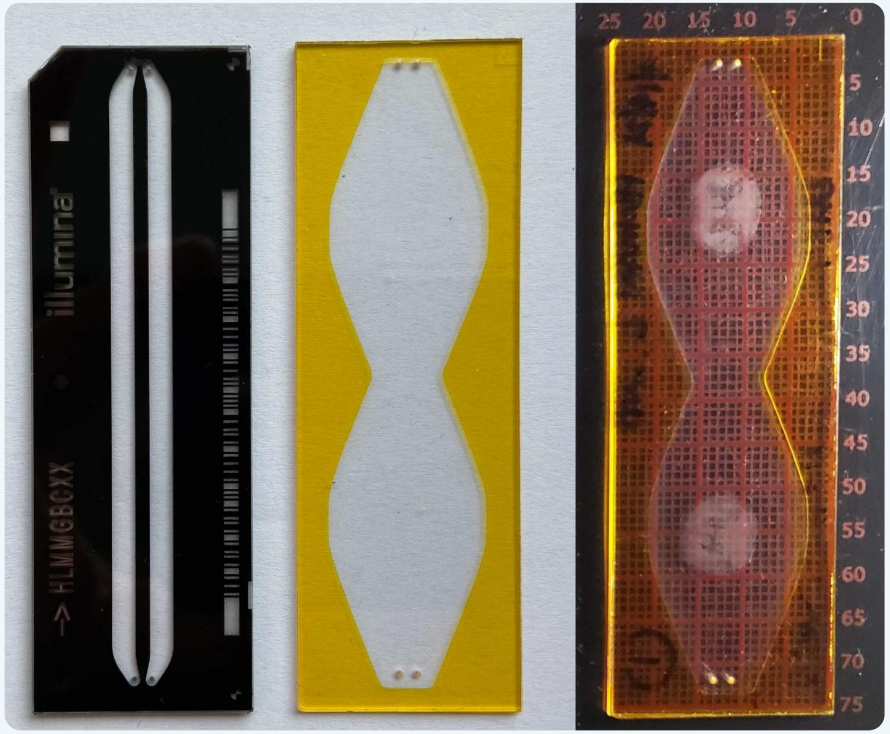
HiSeq X

PySeqX under dev by Clark Lab (Berkeley)

NovaSeq 6000

Planned by the end of the year

Flow cells



Illumina flow cell **PySeq2500 flow cell** **Spinal cord sections**

Available now

HiSeq 2500, 2000 and X. NovaSeq 6000 in development.

Free in a starter kit

We send flow cells to you at no cost.

Buy more from RegenSeq OSE

Additional flow cells are $55 each.

regenseq.github.io/community/request-flowcells

Protocols

Multiplexed Immunofluorescence

Mature 4i and ImmunoSABER

In situ sequencing

Protocol development is led by Maros.

Spatial transcriptomics

Low plex STARmap, starting on RAE-FISH.

Optical pooled screens

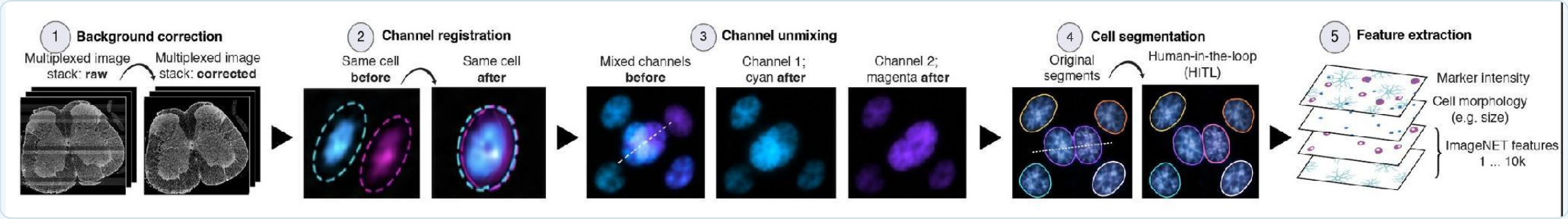
Just getting started.

Beta protocols are coming to our protocols repository, which updates as new ones are ready.

protocols.io/workspaces/regenseq

COMPONENTS · IMAGE PROCESSING

Image processing pipeline



1

Background correction

Corrects uneven lighting across the image stack.

2

Channel registration

Aligns the channels so each cell lines up.

3

Channel unmixing

Separates mixed channels into individual signals.

4

Cell segmentation

Specify custom cellpose models and specify which channels to use for segmentation.

5

Feature extraction

Measures intensity and morphological features and extracts image embeddings.

GET INVOLVED

Who to talk to

Kunal

Vision and code

Overall vision and goals, and the GitHub repositories.



Daniel

Community lead

Getting sequencers, buying flow cells, and training sessions.



Maros

Protocol lead

Manages the protocols repository, especially in situ sequencing.



Sanja

Scientific advisor

Advises on the science of the ecosystem.

GET INVOLVED

How to join in

Contribute code

Help bring up the NovaSeq 6000 by the end of the year, or adapt a model we don't cover yet.

Get a sequencer

We will help you find one to repurpose, and a few at the Genome Center are available.

Order flow cells

Free starter kit for the HiSeq 2500, then additional flow cells at $55 each from the RegenSeq OSE. Daniel can also set up a training session.

Try the protocols

Beta protocols are on the way in the protocols repository.

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